# Assignment No: 8

### Visualization of Real-Time Application Metrics Using Grafana

---

## Aim

To configure Grafana dashboards for visualizing real-time application metrics collected by Prometheus.

## Output

### 1. Verify cluster

```
ronak@ronak:~/devops-task-manager$ minikube start
* Done! kubectl is now configured to use "minikube" cluster
ronak@ronak:~/devops-task-manager$ kubectl get pods
task-manager-deployment-yyyyy-aaaaa   1/1   Running   0   1h
task-manager-deployment-yyyyy-bbbbb   1/1   Running   0   1h
prometheus-deployment-zzzzz-aaaaa     1/1   Running   0   1h
ronak@ronak:~/devops-task-manager$ kubectl get svc
NAME                   TYPE       PORT(S)          AGE
task-manager-service   NodePort   3000:31245/TCP   1h
prometheus-service     NodePort   9090:30090/TCP   1h
```

### 2. Deploy Grafana

```
ronak@ronak:~/devops-task-manager$ kubectl apply -f grafana-deployment.yaml
deployment.apps/grafana-deployment created
ronak@ronak:~/devops-task-manager$ kubectl apply -f grafana-service.yaml
service/grafana-service created
ronak@ronak:~/devops-task-manager$ kubectl get pods
grafana-deployment-wwwww-aaaaa   1/1   Running   0   40s
ronak@ronak:~/devops-task-manager$ minikube service grafana-service
* Opening service default/grafana-service in default browser...
```

### 3. Data source

```
Connections > Data Sources > Add data source > Prometheus
URL: http://prometheus-service:9090
Save & Test
Successfully queried the Prometheus API.
```

### 4. Dashboard panels

```
Dashboard: Student Task Manager Monitoring Dashboard
Application Status    : up{job="task-manager"}                                (Stat)
HTTP Request Rate     : sum(rate(task_manager_http_requests_total[1m]))      (Time series)
Process Memory Usage  : process_resident_memory_bytes                        (Time series, Byte
s)
Node.js Heap Memory   : nodejs_heap_size_used_bytes                          (Gauge)
Refresh interval      : 5s
```

**5. Dashboard output**

```
Student Task Manager Monitoring Dashboard

Application Status
UP

HTTP Request Rate
0.35 requests/second

Process Memory Usage
65 MB

Node.js Heap Memory
8.2 MB

Dashboard Refresh
5 Seconds

Grafana Dashboard Successfully Configured
The dashboard displays real-time application metrics collected by Prometheus.
```

## Conclusion

Grafana was connected to Prometheus and a dashboard was built to visualize application status, request rate and memory usage in real time.